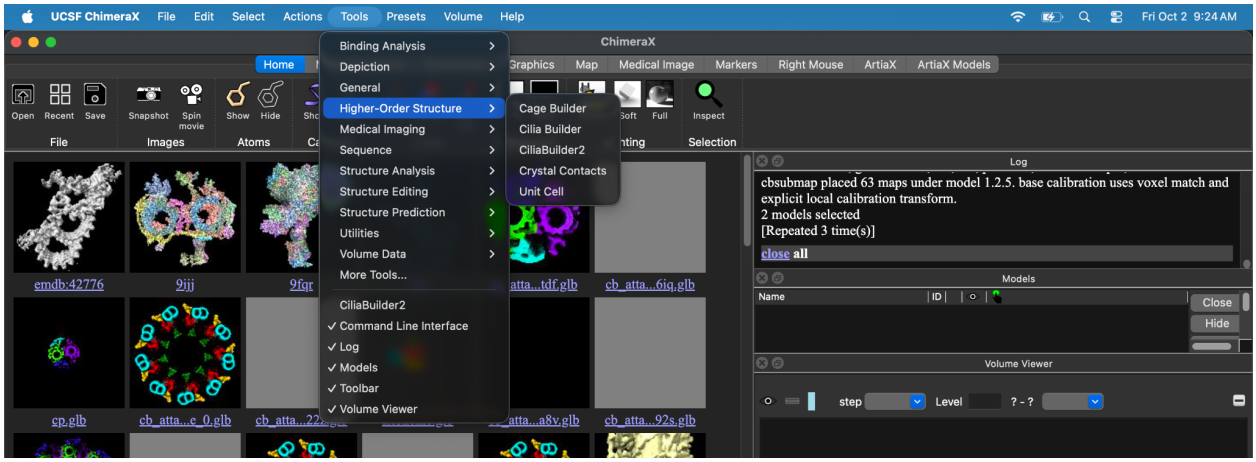
Guidelines for CiliaBuilder2

Last updated: 2026/10/15 (Preliminary)

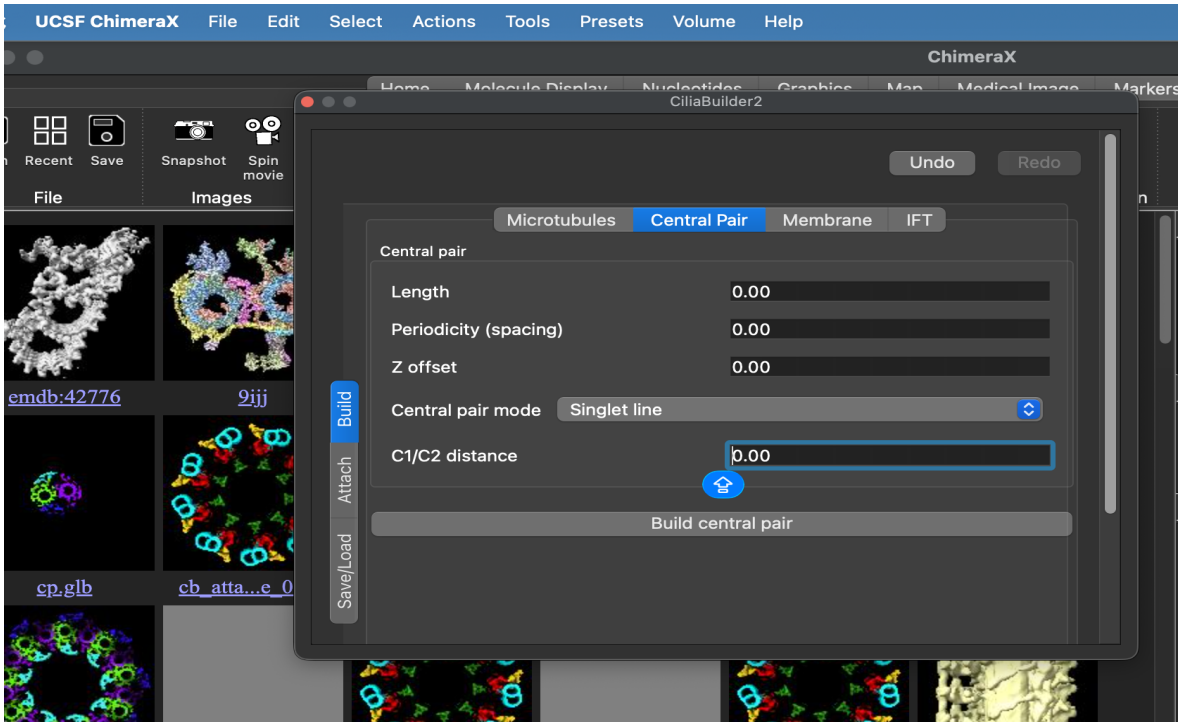
Example 1- Building a simple *Chlamydomonas* cilium

Starting CiliaBuilder2

We will be using the ciliabuilder2.0 to try and create a model of the Chlamydomonas cilia, to do this we need to load the cilia builder interface in ChimeraX, (Tools/Higher Order Structure/Cilia Builder2).



Upon clicking the CiliaBuilder2 interface would open in ChimeraX

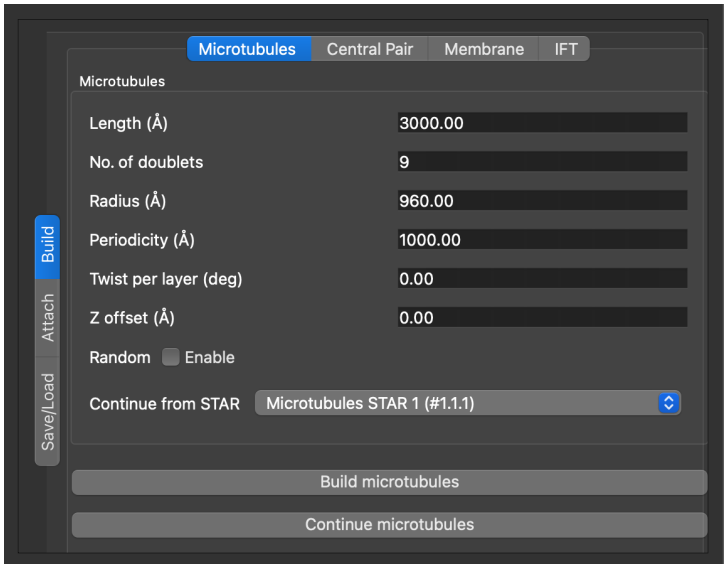


Build the doublet coordinates

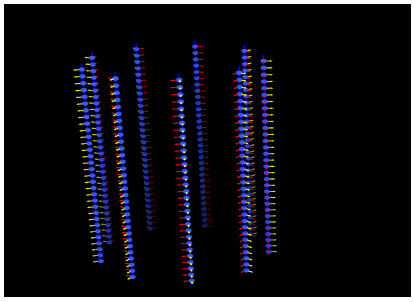
The logic of CiliaBuilder is to **Build → Attach → Save**. For every component, we will **build the coordinates**, then **attach a map**/3D object and finally, **save the JSON file** for the project.

We will start by building coordinates for the microtubules and to do that we will use the build interface mode. The length, periodicity and Z offset can be manipulated by the user and when we want to build microtubules we can click on that on top and when we can do the same thing for the coordinates of CP, membrane and IFT.

NOTE: While we often refer to 96-nm as the repeating unit. The real repeating is 100 nm as tubulin dimer periodicity is not exactly 8-nm. Similarly, the 32 nm repeating unit is 333 Angstrom.



Now we click **Build microtubules** and it should create coordinates in the exact user defined region in the ChimeraX interface as shown below.

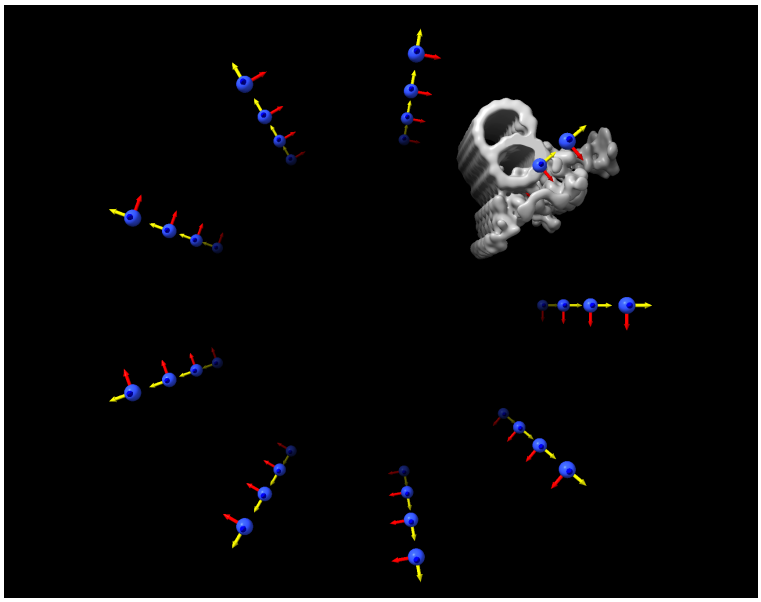


Attach the 96-nm doublet map (EMD-2132)

Before attaching the map, we need to open the map. In ChimeraX, you can type in the Command windows to obtain the EMD-2132, the 96-nm of the Chlamydomonas doublet map.

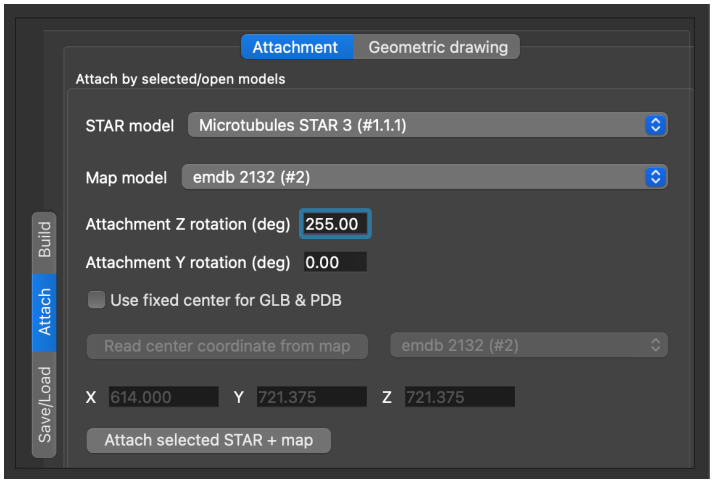
```
open emdb:2132
```

On a Mac or Linux, the map will be downloaded to `~/Downloads/ChimeraX/EMDB` folder.

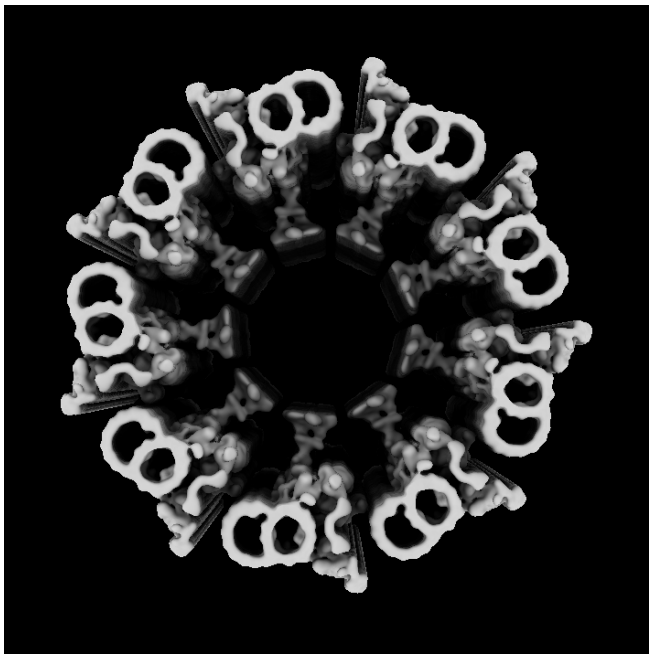


Now let us attach, hover over to the Attach interface, and make sure you are using the appropriate STAR model with appropriate map model (check the numbers of these models on your chimeraX log), here Z and Y can be changed again in an user defined fashion however a default number we are using with this guideline has the following specifications. So, if you use these, one can generate the exact same model.

If in the map, the filament axis is already parallel to the Z axis, then we need only to tweak the Attachment Z rotation to form the cilium model of choice. You can click Attach selected STAR + map many times with different parameters to tweak the angle. In this case:

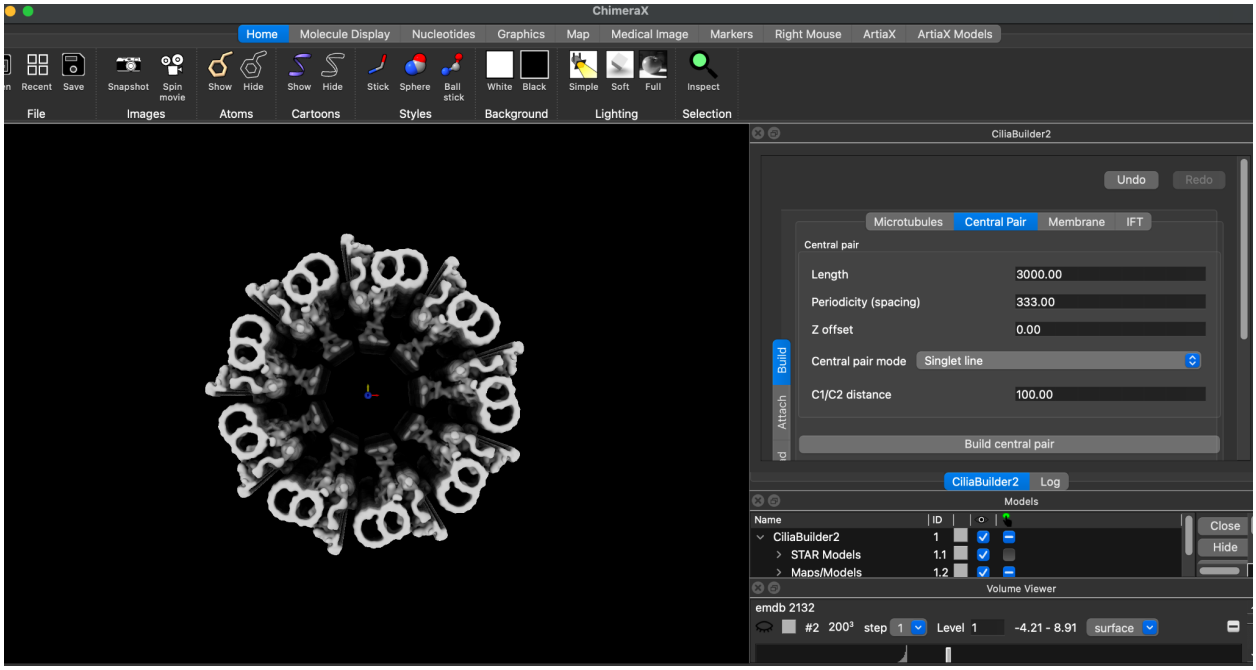


The end result:



Build the central pair coordinates

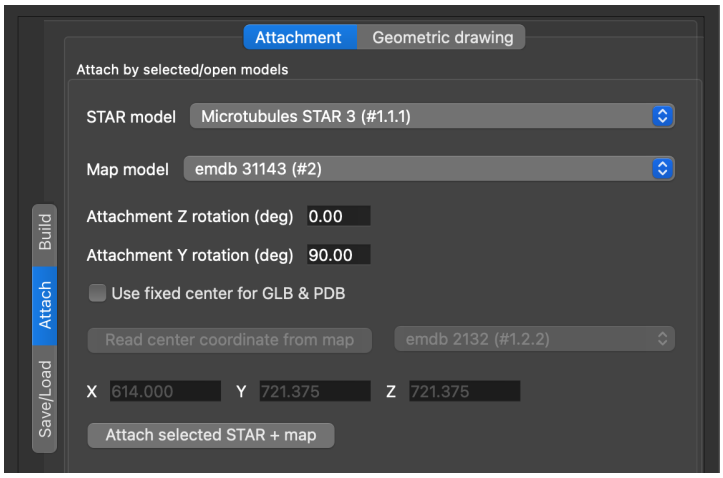
We will again go back to our build interface and place the coordinates for the central pair in this ongoing cilia model. For the same, we are using the following specifications.



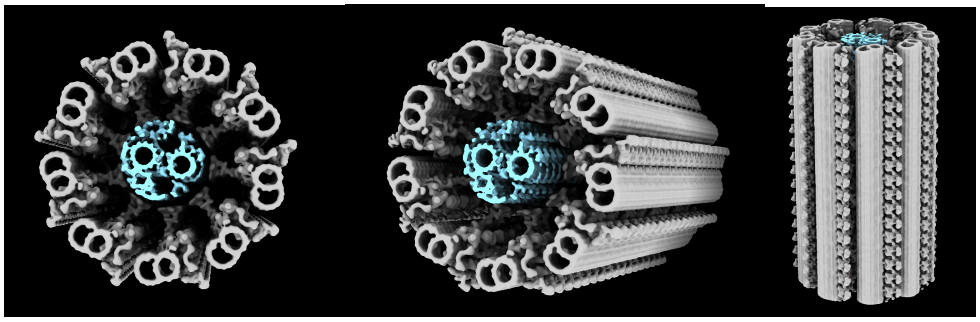
Attach the central pair map (EMD-31143)

Now, in the same way we will download and attach the map EMDB 31143 (Central pair of *Chlamydomonas*) in the appropriate coordinate space. In this case, the filament axis is parallel to Y, therefore, we need to use an Attachment Y rotation angle of 90 degrees.

```
open emdb:31143
```

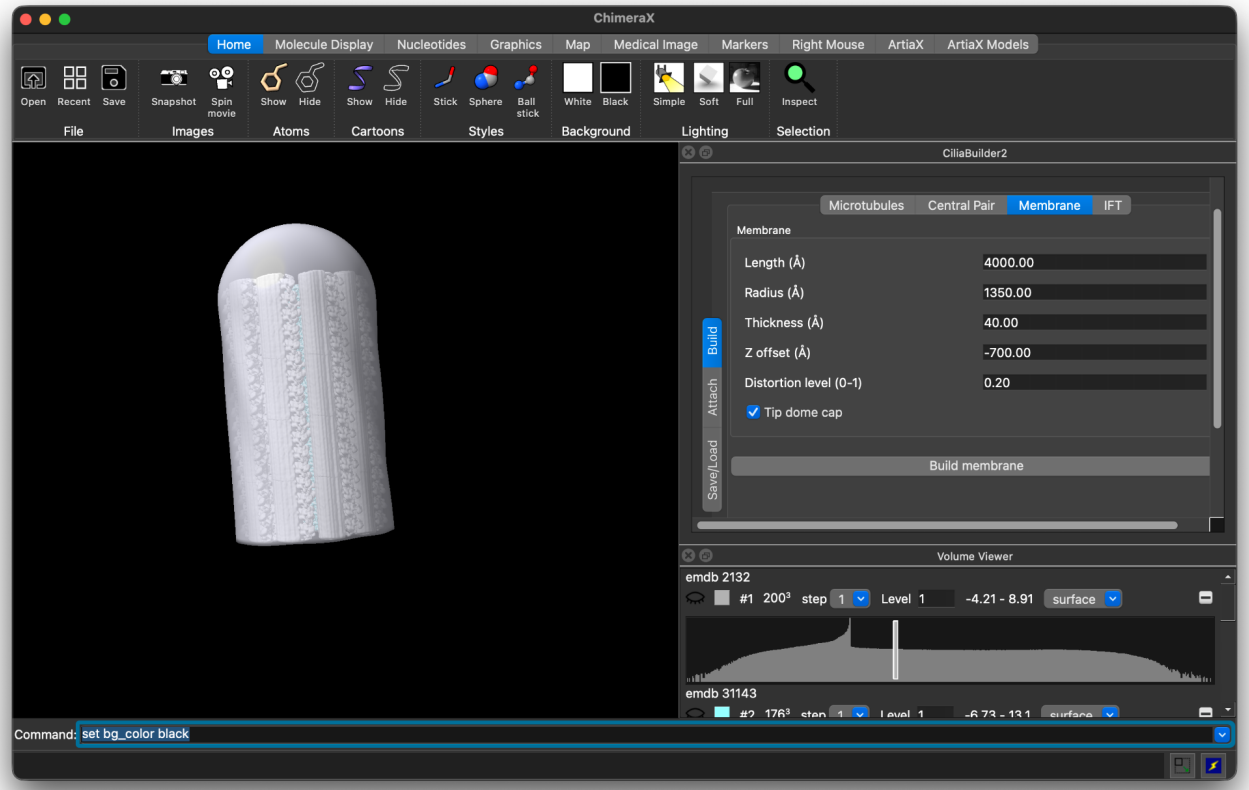


The end result:



Build Membrane

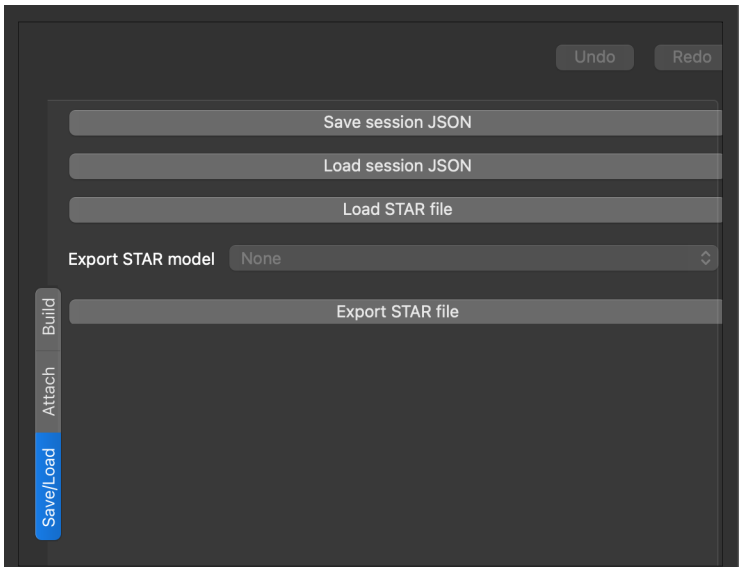
We will extend the rendering capabilities of CiliaBuilder2 by placing a membrane with a domed tip we can use the build interface click on membrane and be sure to check on the tip dome cap and create a membrane, specifications are completely user defined but we will use a following default to accommodate our existing cilia model. We can distort the map to make it a bit less perfect by increasing the value of distortion level to 1.



A note, you might need to adjust the offset to some amount to accommodate the entire length of the microtubule doublet. You can also see we increased the length here much more than it is needed (4000) just to test the versatility of the software.

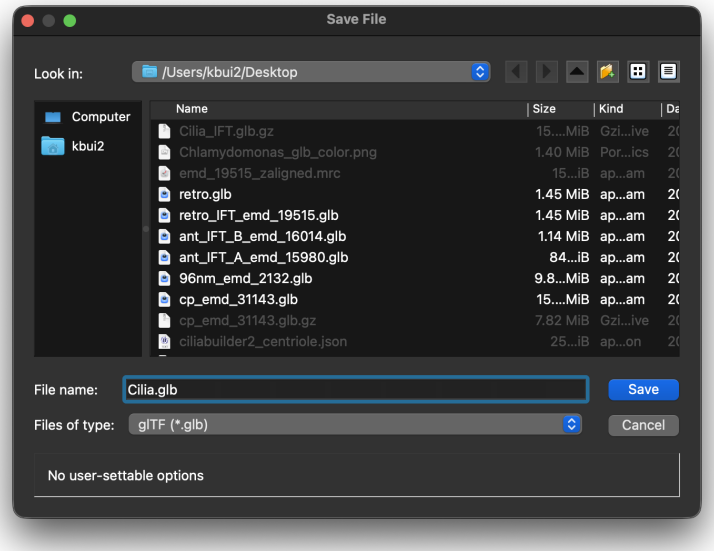
Save JSON session

You can save the session for later loading with a name like CiliaBuilder2_chlamy.json
The JSON file is available on GitHub.
https://github.com/builab/CiliaBuilder2/tree/main/example/1_chlamy



Save GLB of the scene

You can save the scene as a single 3D file using the ChimeraX menu and choose the .gITF file.



After saving e.g. file name Cilia.glb, you can also compress the GLB file using gzip to reduce file size.

```
gzip Cilia.glb
```

The Cilia.glb.gz file can be opened by ChimeraX without any problem.

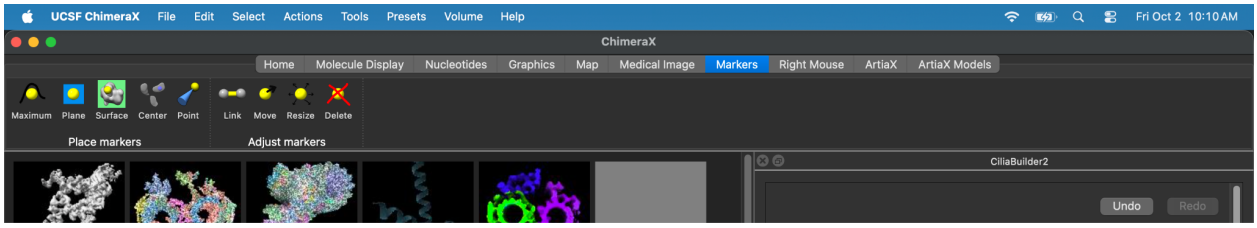
Example 2 :Creating a cilia model using glb file

Here we will use the same EMDB maps 2132 and 31143 to first open these maps in chimeraX color the regions using the marker and color zone tool and then save the colored maps as glb files and then repeat the first example exactly above, only now we will upload the glb files and attach them instead of the original maps. This is to show how user friendly and versatile the software is and every component can be colored and specific regions highlighted in an user defined fashion.

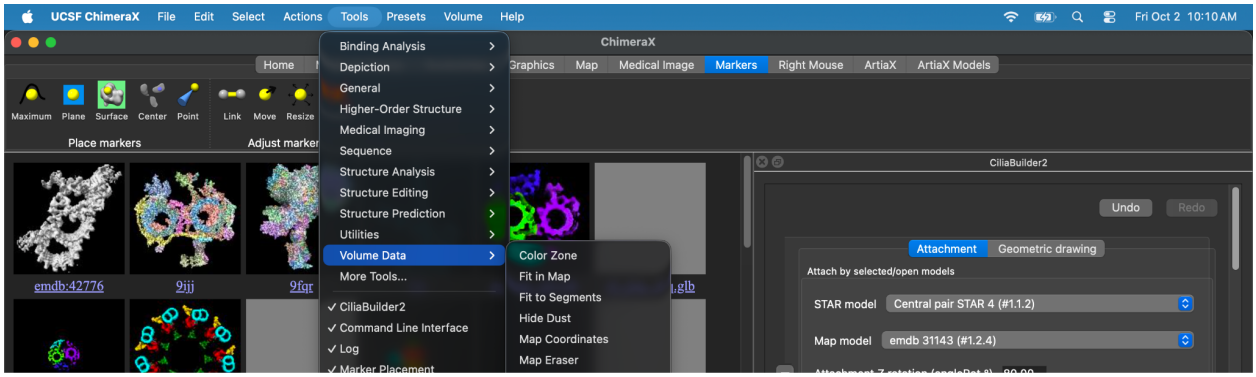
NOTE: All the GLB files are available in GitHub if you want to quickly skip to the Build & Attach step.
https://github.com/builab/CiliaBuilder2/tree/main/example/2_chlamy_glb

Color the doublet map (EMD-2132)

You can use the markers option in ChimeraX and then use the surface icon to mark specific region on your density map and then,refer-
https://www.cgl.ucsf.edu/chimerax/docs/user/tools/markerplacement.html



Go to Color Zone from Tools- and then Volume Data and then color all the regions appropriately
Refer https://www.cgl.ucsf.edu/chimera/docs/ContributedSoftware/colorzone/colorzone.html



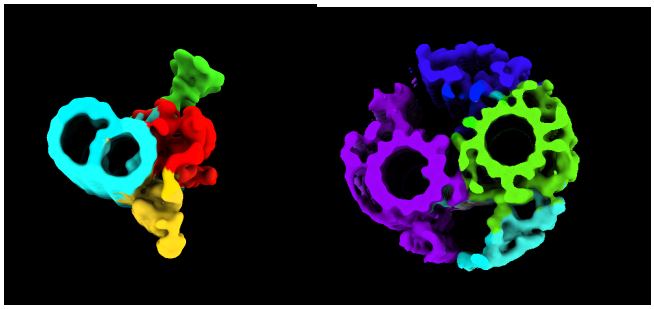
Before saving as GLB, untick the markerset for viewing.
Now we will save our glb files, in the command interface type,

```
save ~/Desktop/96nm_emd_2132.glb center false
```

With the option "center false", the glb will be opened in exactly the same place as the map. Open the GLB file to check this before continuing.

Similarly, color the central pair map in a separate session and save:

```
save ~/Desktop/cp_emd_31143.glb center false
```



Something like shown above

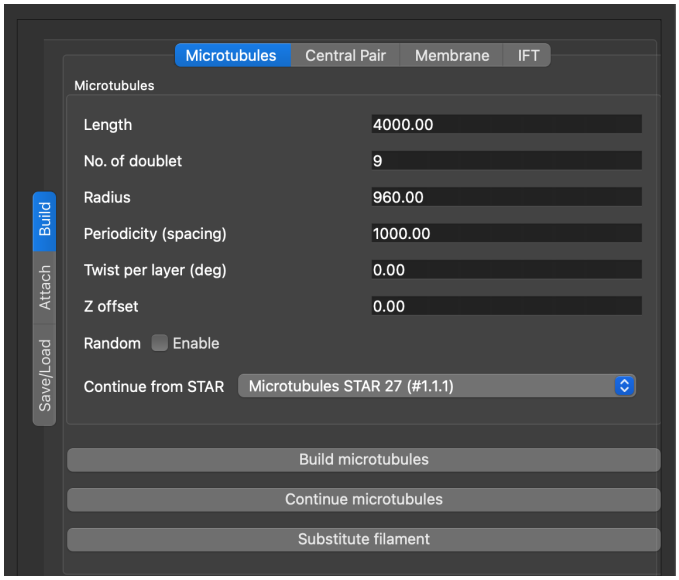
If you are using Linux or Mac, you can compress the file and use it to save space. ChimeraX can read the compressed file.

```
gzip 96nm_emd_2132.glb
gzip cp_emd_31143.glb
```

You will get 96nm_emd_2132.glb.gz and cp_emd_31443.glb.gz file. You can also download these files from GitHub. https://github.com/builab/CiliaBuilder2/tree/main/example/2_chlamy_glb

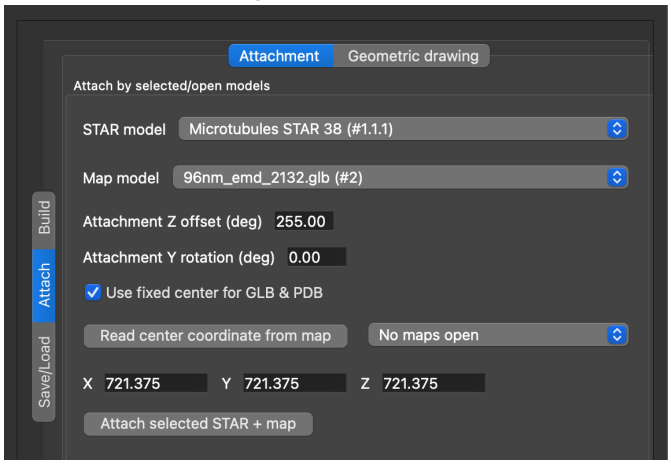
Build the Doublet Coordinate

Now open the CiliaBuilder2 interface and follow the previous steps from the first Chlamydomonas example and you should end up with a model of cilia like shown below. (Only change the length to 4000 Angstrom).



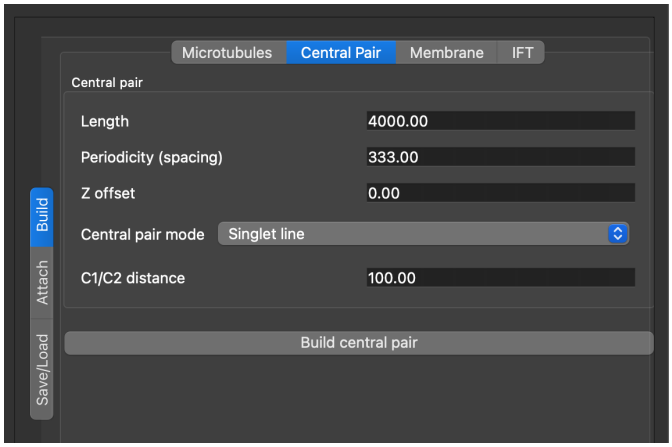
Attach the doublet GLB file

Open the 96nm_emd_2132.glb.gz
You can attach it with the following parameters. If you want to know the center of rotation, you can load the emd_2132 map and then press “Read center coordinate from map” with emd_2132 chosen. In this case, the origin is 721.375, 721.375, 721.375



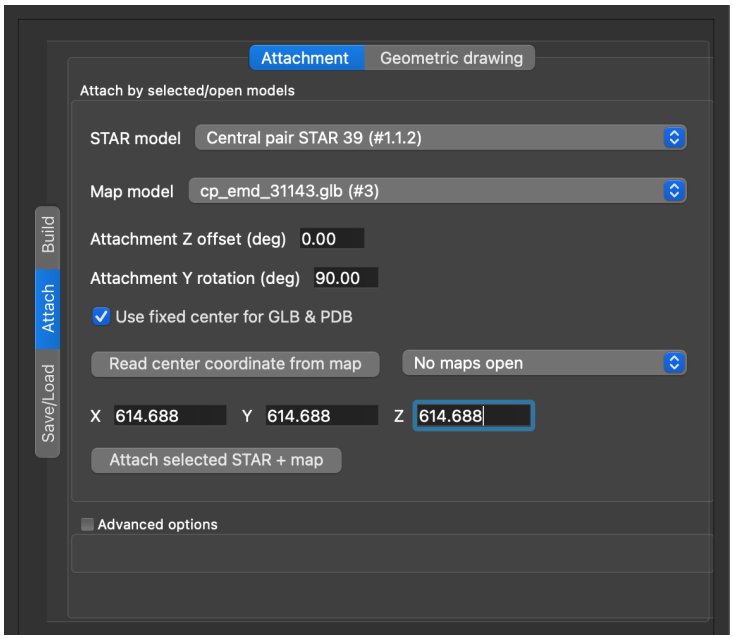
Build the Central Pair Coordinate

We build the central pair coordinate with the following parameters.

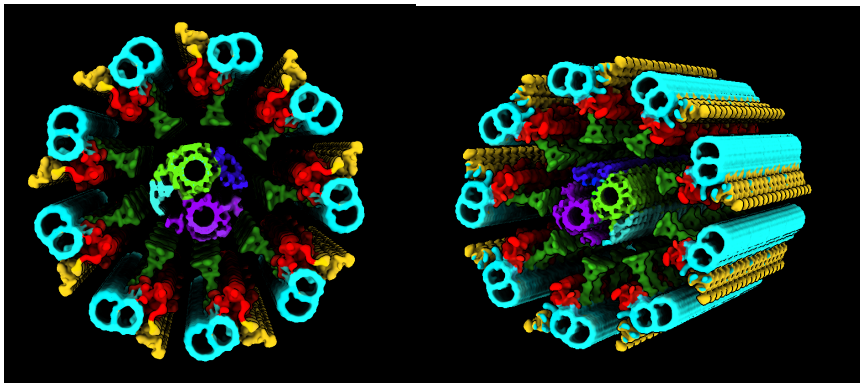


Attach the central pair GLB file

Open the cp_emd_31143.glb.gz
You can attach it with the following parameters. If you want to know the center of rotation, you can load the emd_2132 map and then press “Read center coordinate from map” with emd_31143 chosen. In this case, the origin is 614.688, 614.688, 614.688

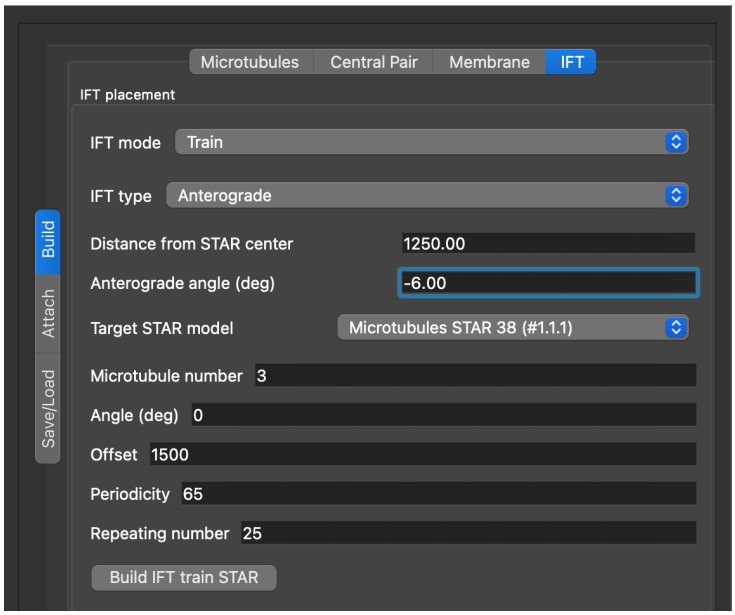


End results



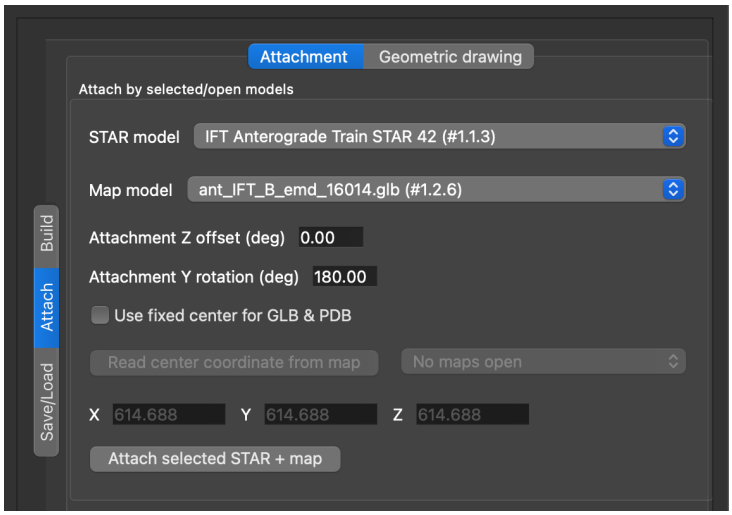
Build the Anterograde IFT-B Coordinate (EMD-16014)

Now we can build even a bit more.
For IFT, we can choose which doublet to run on, with a distance a bit further from the center than the doublet. And to adjust the location of the IFT on the PF of the doublet microtubule, we change the Anterograde angle.
We can choose the offset (how far from 0), periodicity in Angstrom and Repeating number.



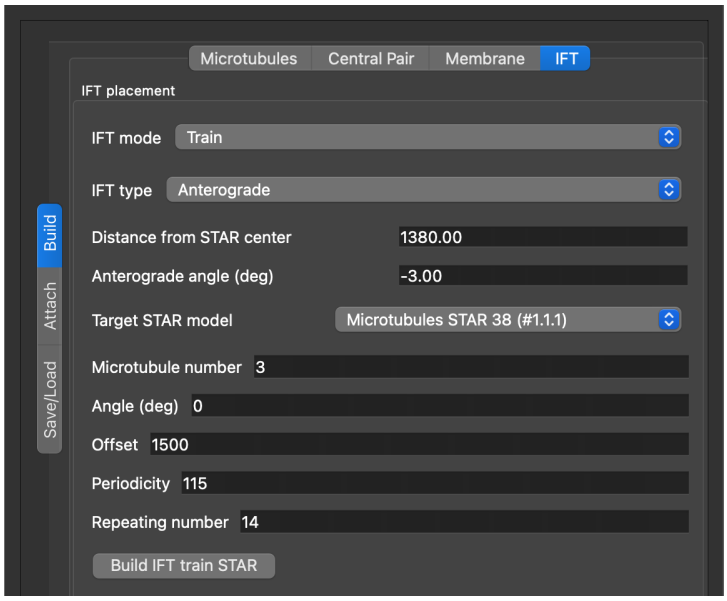
Attach the Anterograde IFT-B GLB file (EMD-16014)

Now, open and attach the ant_IFT_B_emd_16014.glb.gz file with the following parameters. In this case, we don't need to read the fixed center as we don't need to align it with the existing map.



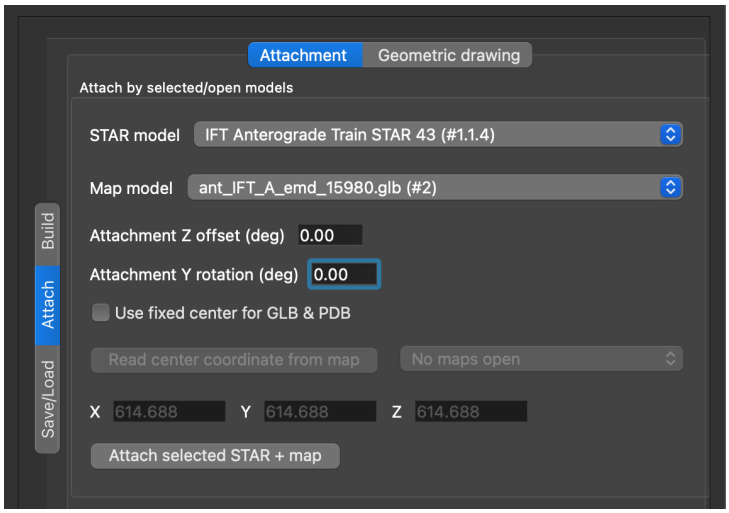
Build the Anterograde IFT-A Coordinate (EMD-15080)

For the IFT B component, we need it to be a bit further from the center so it is on top of IFT-B. The anterograde angle is similar to IFT-B.



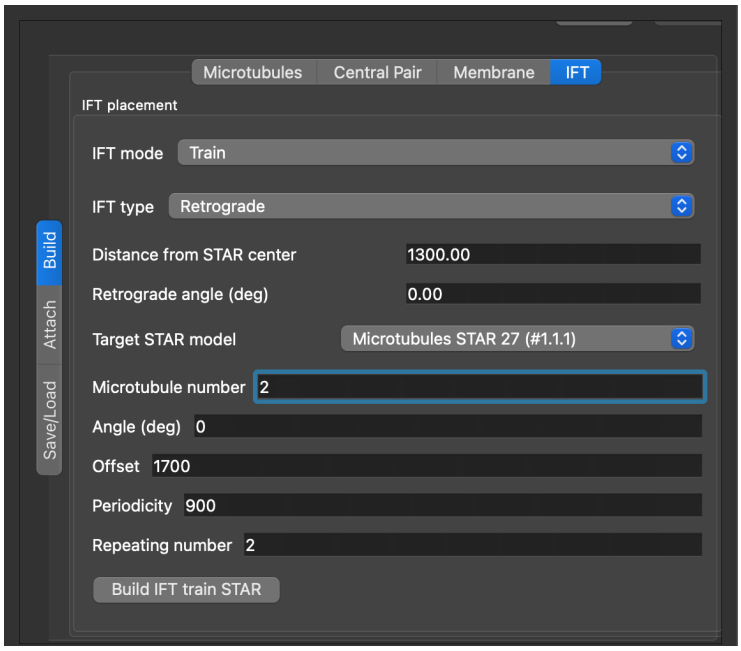
Attach the Anterograde IFT-A GLB file (EMD-15080)

Now, open and attach the ant_IFT_A_emd_15080.glb.gz file with the following parameters. As the original map EMD-15080 is not parallel to Y or Z axis, we have aligned it parallel to Z using the script z_align_emd15980.cxc from GitHub before coloring and saving the GLB file.



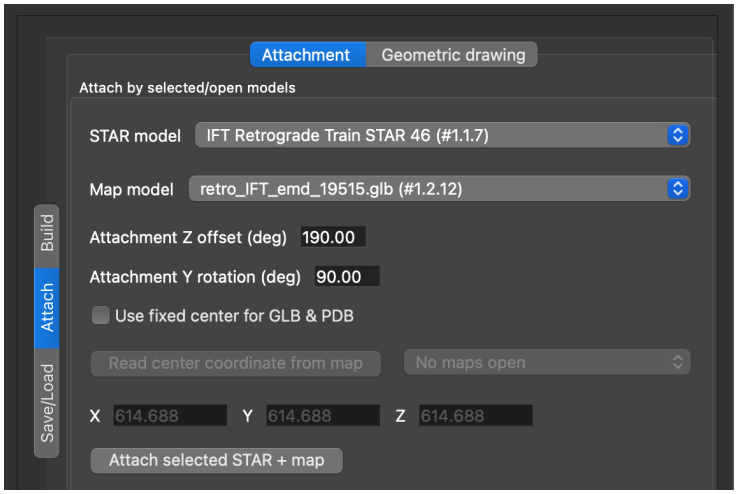
Build the Retrograde IFT Coordinate (EMD-19515)

Similar to Anterograde Train,, we can build retrograde trains by choosing which doublet to run on, with a distance a bit further from the center than the doublet and retrograde angle.

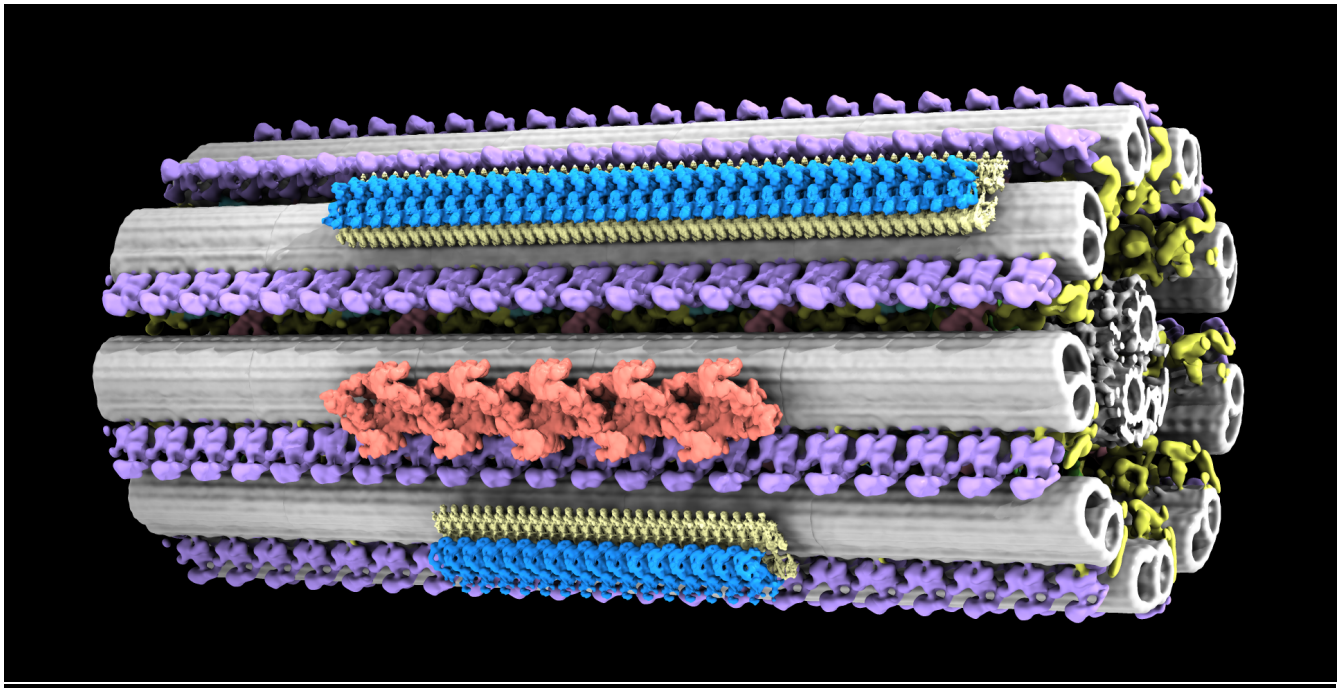


Attach the Retrograde IFT GLB file (EMD-19515)

Now, open and attach the retro_IFT_emd_19515.glb.gz file with the following parameters. As the original map EMD-19515 is not parallel to Y or Z axis, we have aligned it parallel to Z using the script z_align_emd19515.cxc from GitHub before coloring and saving the GLB file.



End result



For this complicated session, please save a JSON file. All the GLB files and the JSON for this example can be downloaded from GitHub and recreated by the Load JSON file feature.

Example 3: Bovine Sperm Model using large PDB files

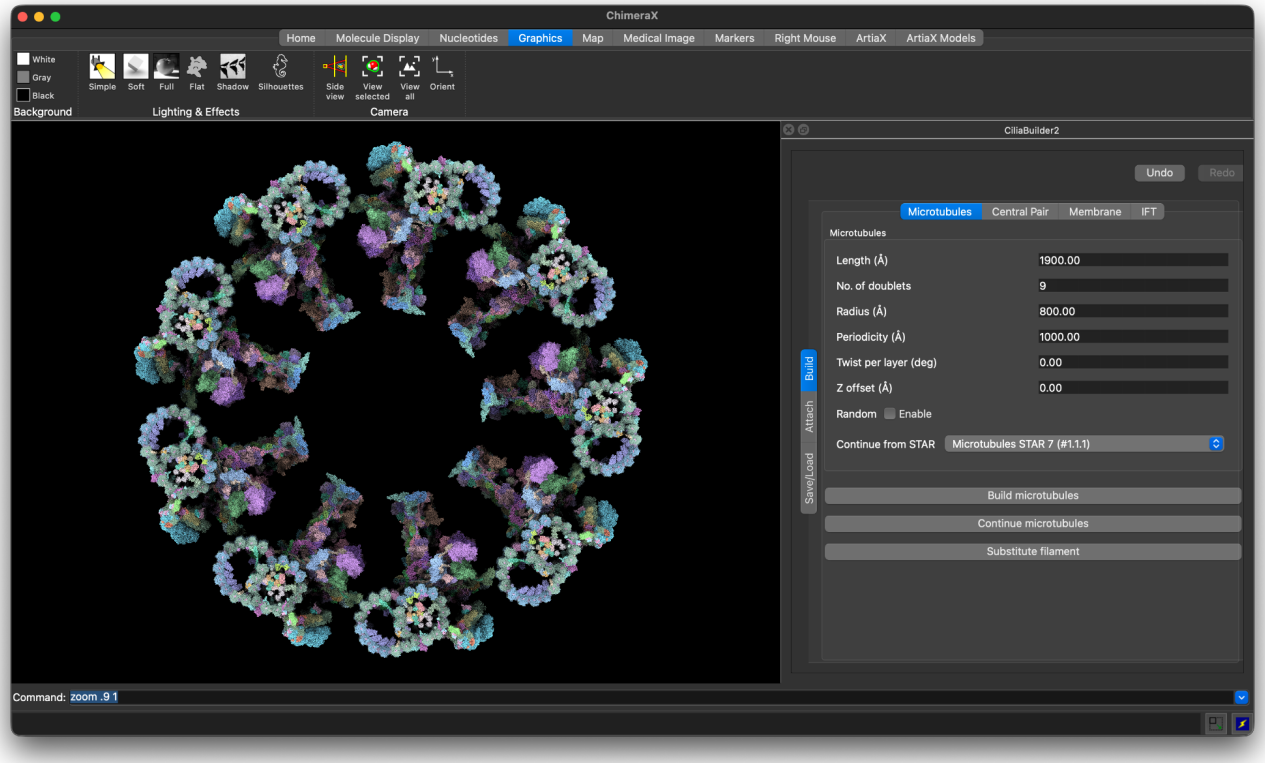
We will use models now to create a ciliary structure of bovine sperm, this might take some time because of restrictions in CPU, GPU and RAM. Nonetheless, given millions of atoms are modelled here, **CiliaBuilder2** does a decent job of creating the same.

Build Microtubule Coordinate & Attach

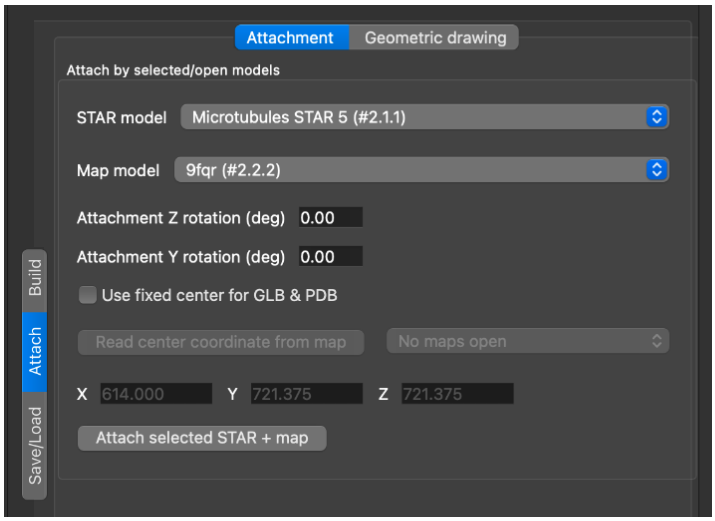
We will first open the mmCIF files directly into our ChimeraX interface using the command prompt. First we will load the DMT 9fqr- 96 nm repeat from the axonemal DMT of bovine sperm.

```
open pdb:9fqr
```

Following that we will add the coordinates using the following specifications. Here we use Radius of 800 Angstrom only as for PDB, the center of rotation by default is the center of bounding boxes, not center of the volume like a mask.



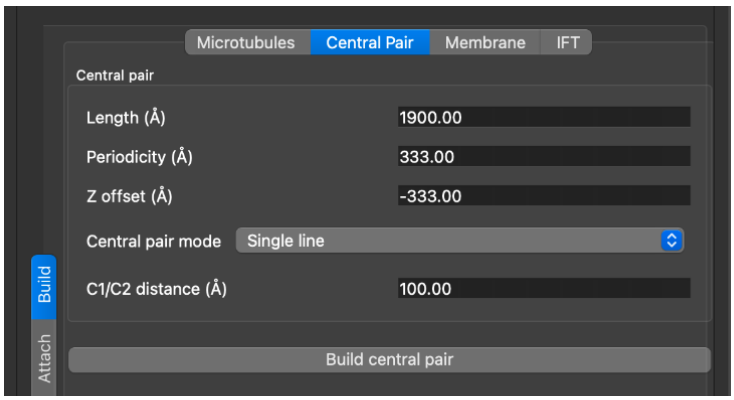
Attachment specification:
Now we will attach the DMTs along the placed coordinates using the attach mode.



We have to be mindful about the angle of rotation to create the most biologically plausible model of the cilia; nonetheless, these parameters can be changed by the user.

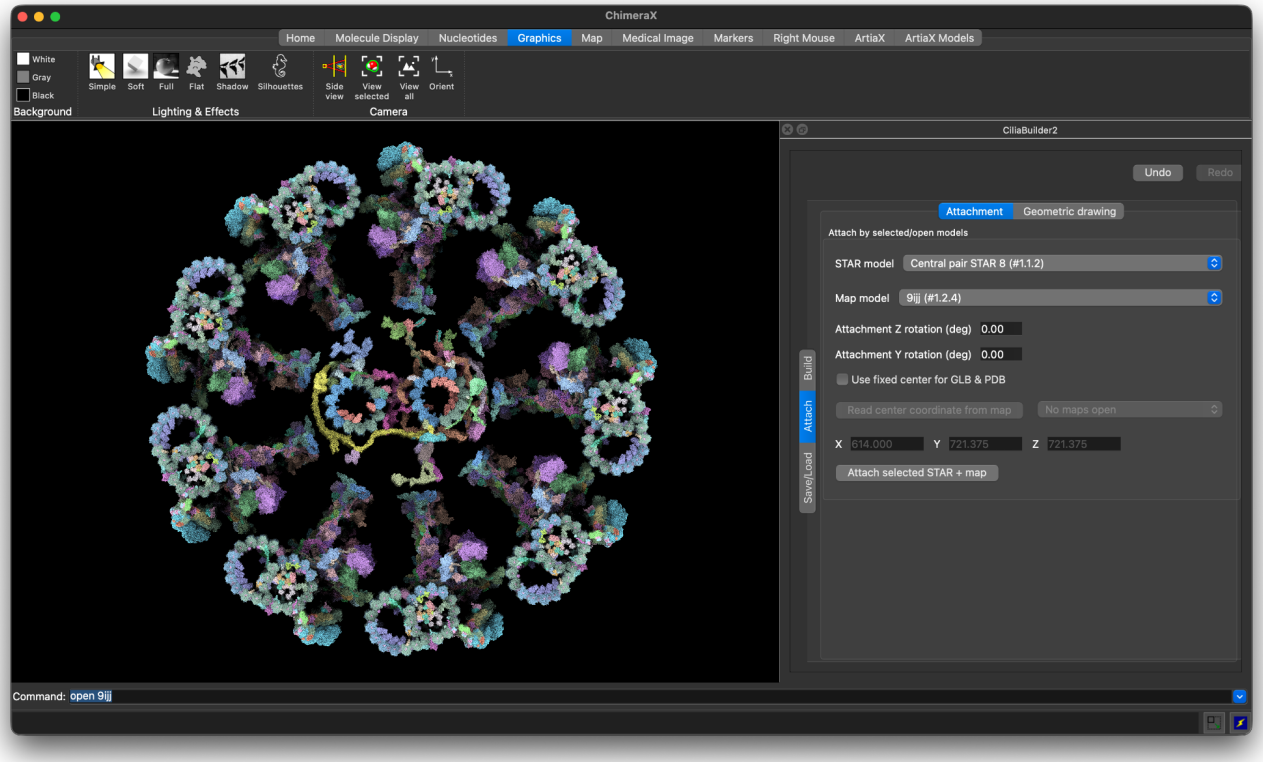
Build Central Pair Coordinate & Attach

Next we will build coordinates for the CP and attach the appropriate model pdb 9ijj

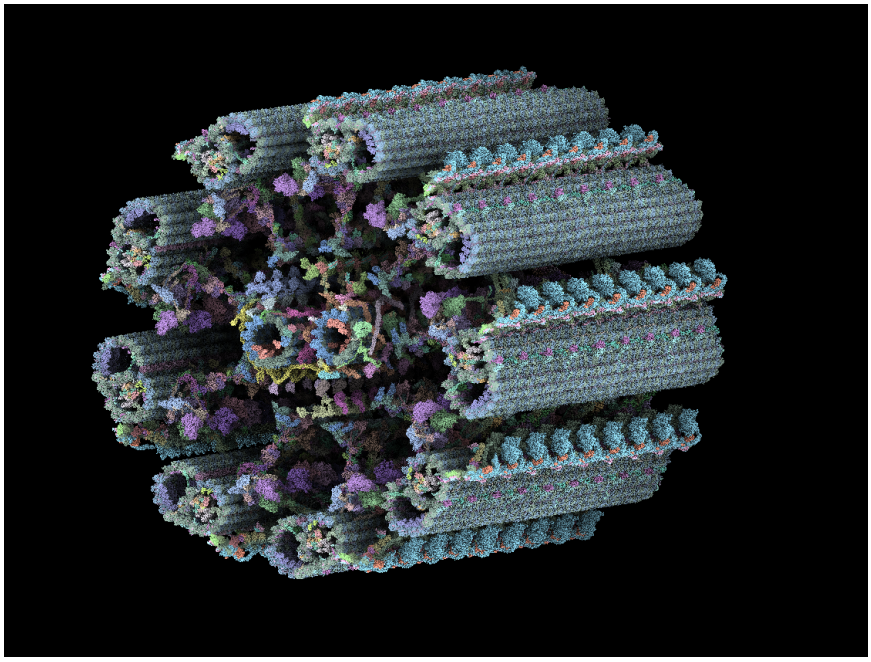



```
open pdb:9ijj
```

Then in a similar fashion, we will attach the central pair STAR file to the appropriately loaded 9ijj.



End result



The structure here should look something like this, parameters here also can be changed in user defined manner.

Example 4- Building a Cartwheel Docked Basal Body Model

In this example, we will build a centriole using available maps from Tetrahymena centriole (EMD-42776, EMD-42777, EMD-42778) and the cartwheel from Trichonympha (EMD-2329). To make the process easy, we provide you with the colored and aligned GLB files for CiliaBuilder2 already from GitHub.
https://github.com/builab/CiliaBuilder2/tree/main/example/4_centriole

Adjust map orientation

There are two problems with the maps:

- (i) The triplet microtubules in the three maps are unfortunately not aligned to the Z axis, making it difficult or even not possible to adjust when attached.
- (ii) The microtubules in the 3 maps are not aligned to each other, not allowing seamless continuity.

So, we have made 3 scripts to do both of this.
Open ChimeraX, then open z_align_emd42776.cxc. This script will download EMD-42776, align to the Z axis and the map.
Close the session, open z_align_emd42777.cxc. It will align EMD-42776 to the Z axis and then align EMD-42777 on EMD-42776.
Close the session, open z_align_emd42778.cxc. It will align EMD-42776 to the Z axis and then align EMD-42778 on EMD-42776.

All the 3 maps are available in the Example:
emd_42776_zaligned.mrc, emd_42777_zaligned.mrc, emd_42778_zaligned.mrc

Make colored GLB files

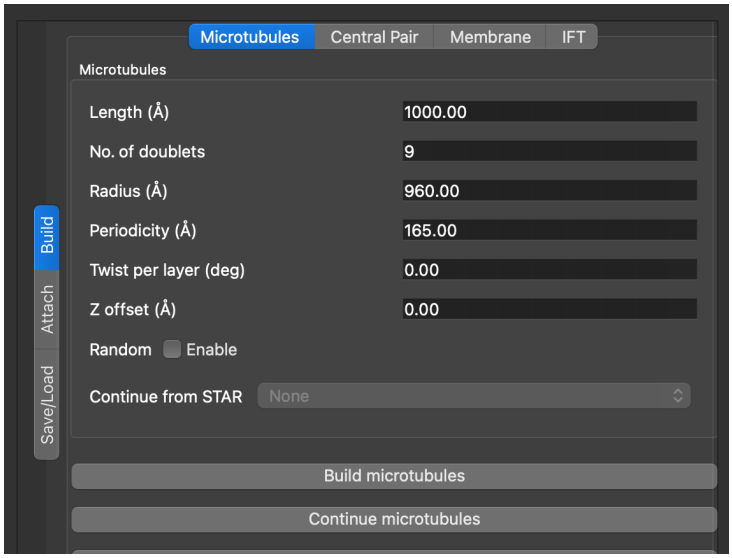
We colored all none-tubulin densities from the maps and saved the corresponding map as GLB file with “center false” option (See Example 2). You can just download the GLB files:

`proximal_emd_42776.glb.gz`
`core_emd_42777.glb.gz`
`distal_emd_42778.glb.gz`

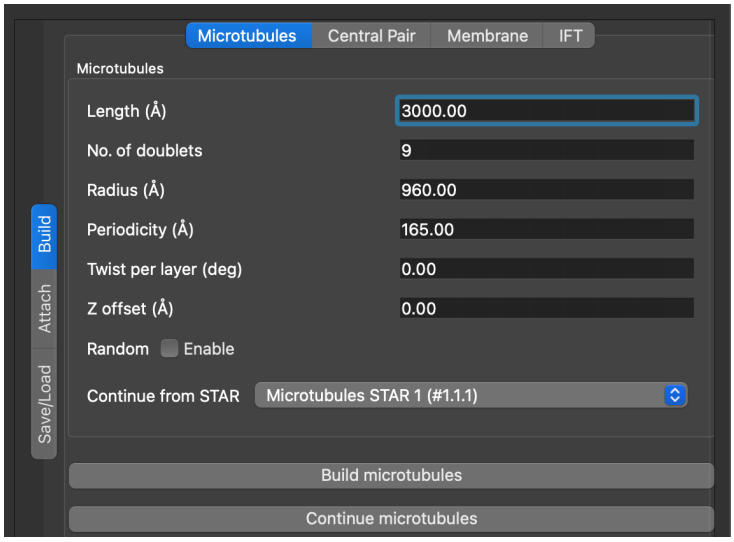
Build Centriole Coordinates

Here we will build a model of centriole with 3 regions. Proximal: 100 nm, Core: 300 nm and Distal: 100 nm using Build Microtubule and Continue Microtubule. “Continue Microtubule” remembers the end point of the previous coordinate and just extends further.

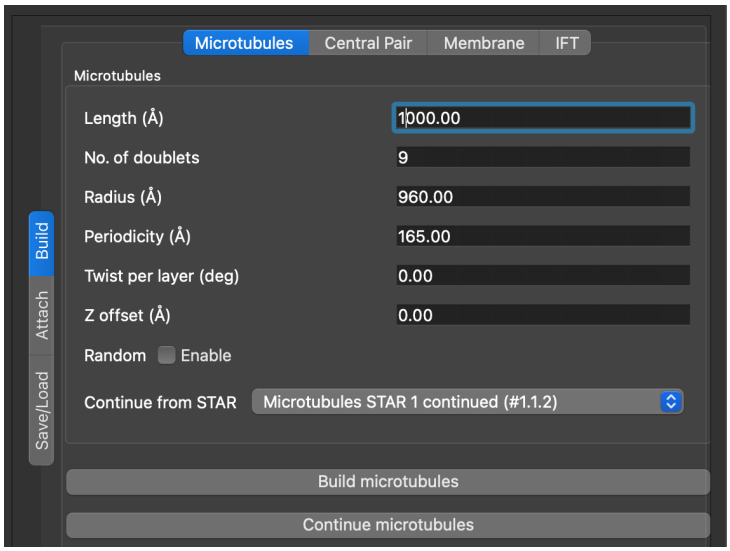
Build Proximal



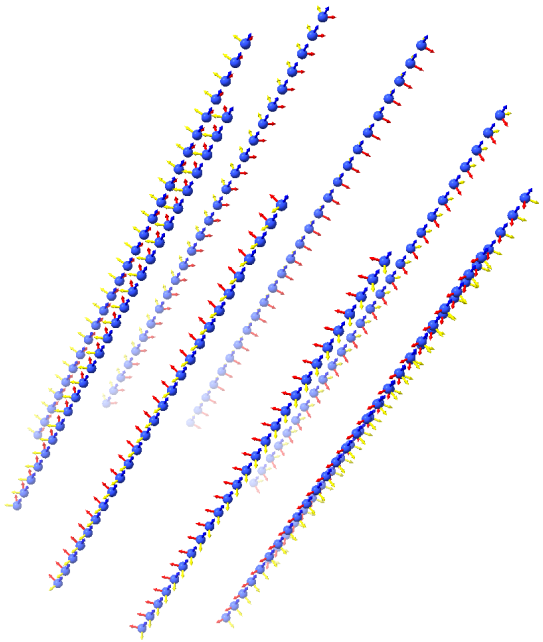
Continue Microtubule for the core region



Continue Microtubule for the distal region



End results



Attach Centriole GLB files

Now we attached the centriole GLB files

Open

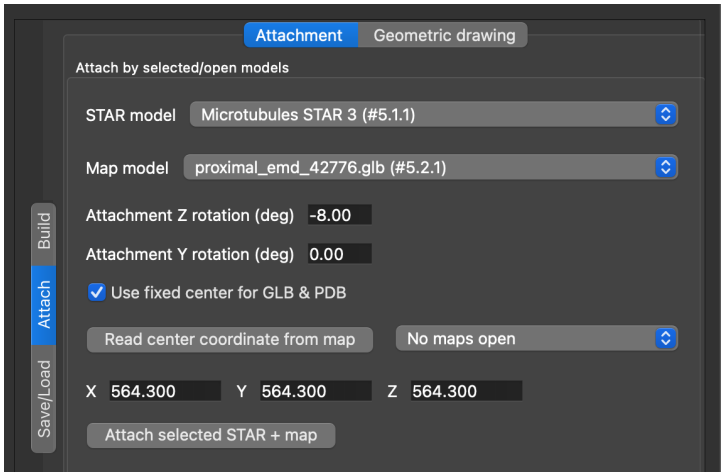
`proximal_emd_42776.glb.gz`

`core_emd_42777.glb.gz`

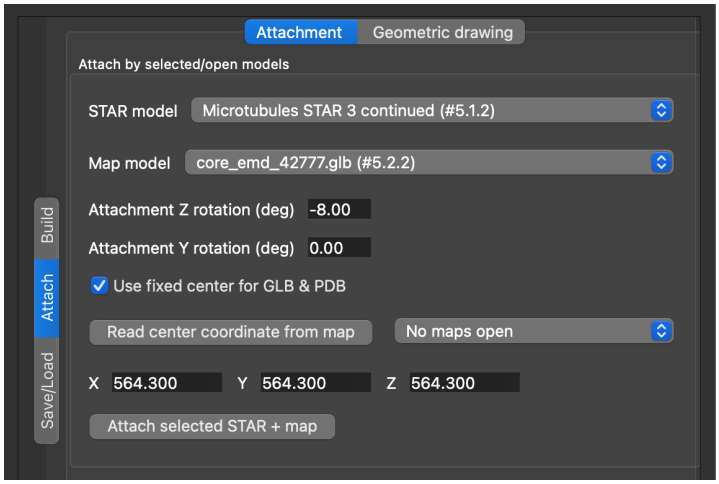
`distal_emd_42778.glb.gz`

Attach each of the map with the following parameters

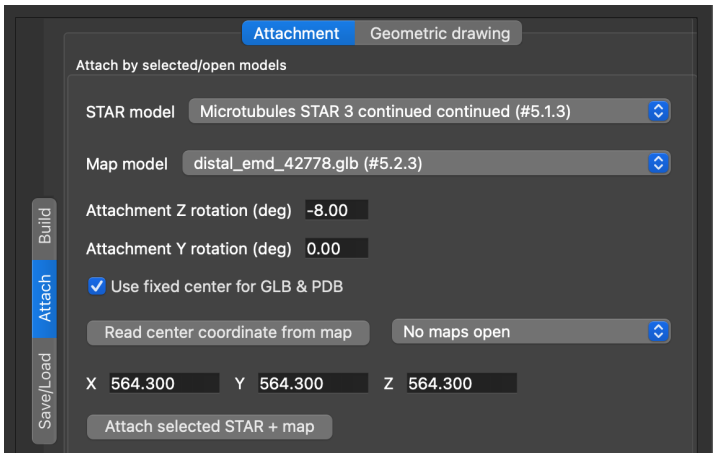
Proximal



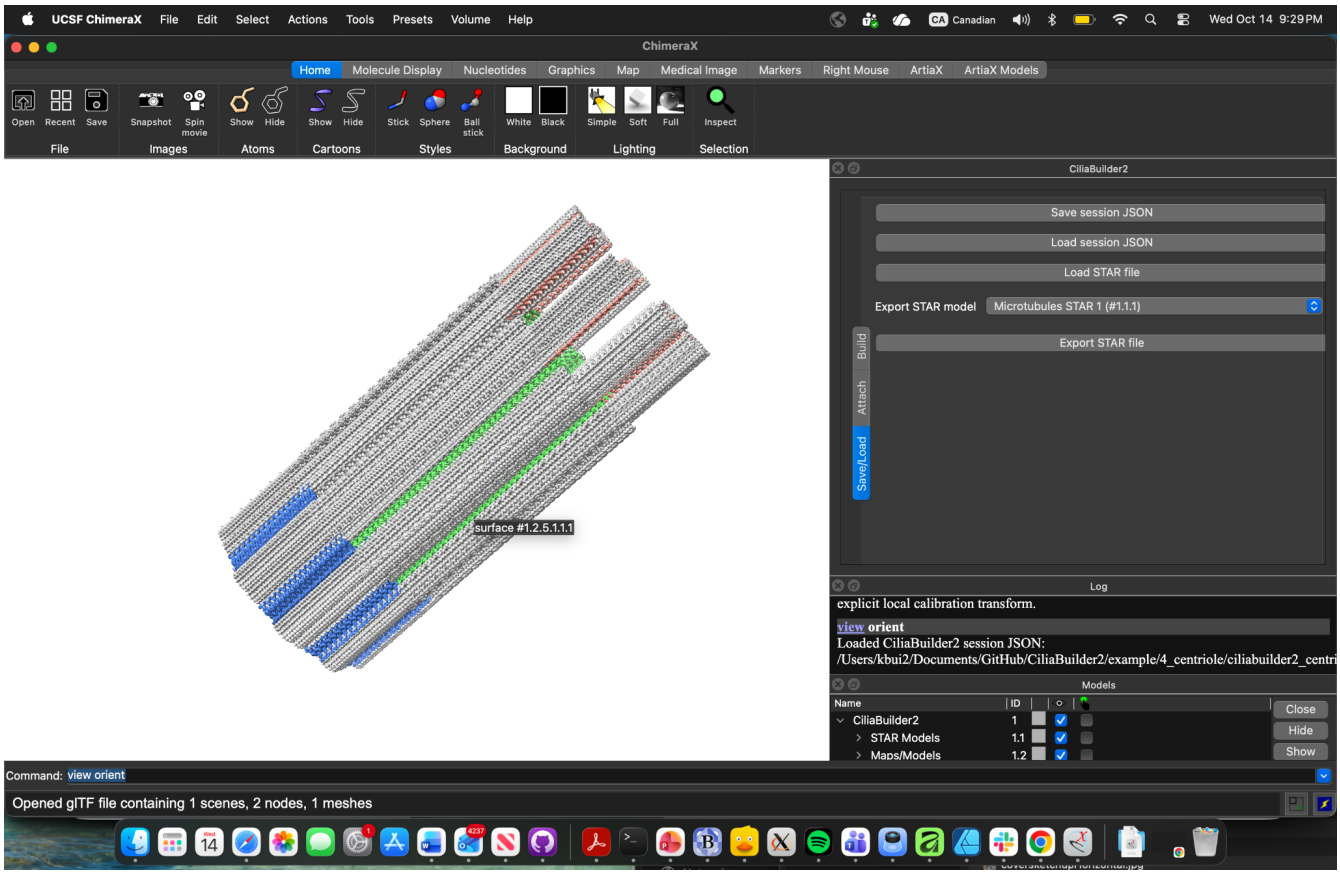
Core



Distal

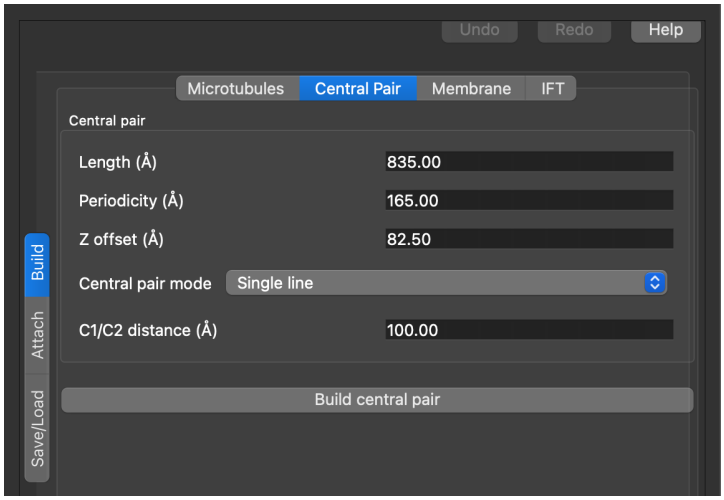


We got this after



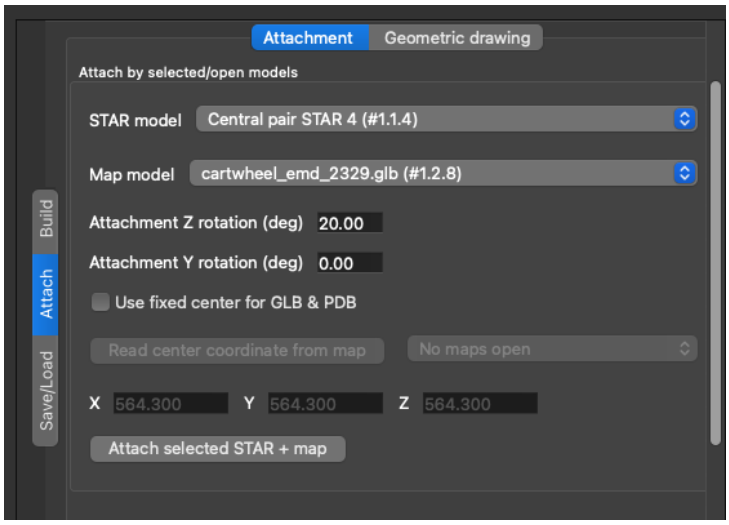
Build the cartwheel coordinate

Build the cartwheel in the proximal region using the Central Pair menu.



Attach the cartwheel coordinate

Open the cartwheel GLB file: cartwheel_emd_2329.glb.gz
Attach with the following parameter



Final result

